

RAPID3 (Routine Assessment of Patient Index Data 3), a Rheumatoid Arthritis Index Without Formal Joint Counts for Routine Care: Proposed Severity Categories Compared to Disease Activity Score and Clinical Disease Activity Index Categories

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ABSTRACT. *Objective.* To compare 4 categories (high, moderate, and low severity, and near-remission) of RAPID3 (Routine Assessment of Patient Index Data 3), an index without formal joint counts, which is scored in < 10 seconds to 4 categories of the Disease Activity Score (DAS28) and Clinical Disease Activity Index (CDAI) in patients with rheumatoid arthritis (RA).

Methods. All patients complete a Multidimensional Health Assessment Questionnaire (MDHAQ) at each visit. A physician/assessor 28-joint count and erythrocyte sedimentation rate (ESR) were completed in 285 patients with RA in usual care by 3 rheumatologists to score DAS28, CDAI, and RAPID3. RAPID3 includes the 3 MDHAQ patient self-report RA Core Data Set measures for physical function, pain, and patient global estimate. Proposed RAPID3 (range 0–10) severity categories of high (> 4), moderate (2.01–4), low (1.01–2), and near-remission (≤ 1) were compared to DAS (0–10) activity categories of high (> 5.1), moderate (3.21–5.1), low (2.61–3.2), and remission (≤ 2.6), and CDAI (0–76) categories of > 22, 10.1–22.0, 2.9–10.0, and ≤ 2.8 . Additional RAPID scores, which add to RAPID3 a physician/assessor or patient self-report joint count and/or assessor global estimate, were also analyzed. Statistical significance was analyzed using Spearman correlations, cross-tabulations, and kappa statistics.

Results. All RAPID scores were correlated significantly with DAS28 and CDAI ($\rho > 0.65$, $p < 0.001$). Overall, 78%–84% of patients who met DAS28 or CDAI moderate/high activity criteria met similar RAPID severity criteria, and 68%–77% who met DAS28 or CDAI remission/low activity criteria also met similar RAPID criteria. RAPID3 was as informative as other indices.

Conclusion. RAPID3 provides a feasible, informative quantitative index for busy clinical settings. (First Release Sept 15 2008; J Rheumatol 2008;35:2136–47; doi:10.3899/jrheum.080182)

Key Indexing Terms:

RHEUMATOID ARTHRITIS

ROUTINE ASSESSMENT OF PATIENT INDEX DATA

MULTIDIMENSIONAL HEALTH ASSESSMENT QUESTIONNAIRE

REMISSION

SEVERITY

Quantitative assessment has advanced therapies for patients with rheumatoid arthritis (RA) over the last 2 decades. “Tight control” according to a Disease Activity Score 28 (DAS28)^{1–3} is associated with significantly better outcomes than usual nonquantitative care of RA^{4–9}. However, most RA patient care at this time generally is guided only by a care-

ful, but nonquantitative, history and physical examination. The only quantitative measures included are laboratory tests, which often are not informative and/or not available at the time of the visit¹⁰. A formal quantitative joint count is not performed at most visits¹¹, so that DAS28 or Clinical Disease Activity Index (CDAI)¹² are not available. “Documentation” of patient status and changes over time generally is available only from “gestalt” clinical impressions.

Each of 7 RA Core Data Set measures^{13,14} have similar relative efficiencies to distinguish active from control treatments in clinical trials^{15,16}. Therefore, indices composed of only 3 or 4 Core Data Set measures such as the DAS28¹⁷ or of only the 3 patient-reported Core Data Set measures (physical function, pain, patient global estimate) without joint counts, distinguish active from control treatments in clinical trials (of leflunomide^{18,19}, methotrexate^{18,19}, adalimumab²⁰, and abatacept²¹) at levels similar to American

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College of Rheumatology (ACR) criteria²² and to one another. Scores on the “patient-only” indices are correlated significantly with DAS28 in these clinical trials^{18–21} as well as in clinical settings²³.

One “patient-only” index, termed the “routine assessment of patient index data 3” (RAPID3), can be scored in fewer than 10 seconds on a multidimensional Health Assessment Questionnaire (MDHAQ)^{24,25} (Figure 1), compared to about 42 seconds for a standard HAQ, and 90 seconds for a quantitative 28-joint count²⁶. The MDHAQ can be completed by most patients in 5–10 minutes prior to seeing a rheumatologist, so that a RAPID3 score is available at the time the patient is seen. A RAPID3 score could provide a pragmatic quantitative index for a rheumatologist to assess, monitor, and document patient status in a busy clinical setting.

We compared proposed categories of high, moderate, and low severity and near-remission for RAPID3 to categories for high, moderate, and low activity and remission for DAS28^{3,27} and CDAI^{28,29} in 285 patients with RA seen by 3 rheumatologists in usual clinical care. We also analyzed additional RAPID scores, which included a joint count by a physician/assessor or by patient self-report³⁰ and/or a physician global estimate, to assess whether inclusion of these data might provide a substantially more informative index than RAPID3.

MATERIALS AND METHODS

Patients. Patients were studied from 3 rheumatology clinical settings of MB, TP, and YY. Patients were seen by MB in a private-practice setting established in 1987, by TP in an academic setting since 1980, and by YY in both private practice and academic settings since 2001. Each patient (with any diagnosis) seen by these rheumatologists completes a version of an MDHAQ^{24,25} at each visit. In addition, 100 consecutive patients with RA at each setting were assessed in usual care according to a “standard protocol to evaluate RA” (SPERA)³¹, which also includes a 28-joint count, laboratory tests, and further clinical assessments of RA (see below).

Patients signed consent for de-identified results to be sent anonymously to a data center at Vanderbilt University (Nashville, Tennessee); data from Vanderbilt University patients were de-identified for this study. The study was approved by the Institutional Review Board for the Protection of Human Subjects at Vanderbilt University, and at the other settings. The 3 settings are the US participants in the international QUEstionnaire in STandard clinical care of RA (QUEST-RA) program³².

MDHAQ questionnaire. The MDHAQ (Figure 1)^{24,25} is a 2-sided, single-sheet instrument, adapted from the standard HAQ³³, designed to facilitate review and scoring by a health professional in a busy clinical setting. Patients complete the MDHAQ while waiting to see the physician, so that scores are available for physician review at the time the patient is seen. Many versions have been developed in response to clinical observations and requests of rheumatologists; all versions are at least 80% identical.

Version R783 of the MDHAQ includes 5 scales on Page 1 (Figure 1A) to assess physical function, psychological distress, pain, patient global estimate, and a self-report joint count on a RA Disease Activity Index (RADAI)^{30,34}. Thirteen items (1a–1m) are queried for 4 responses: without any difficulty (= 0), with some difficulty (= 1), with much difficulty (= 2), and unable to do (= 3), as on the HAQ³³. The first 10 items (1a–1j) are activities, 8 identical to the HAQ, one from each of the 8 HAQ categories, reported as a modified HAQ (MHAQ) in 1983³⁵, as well as 2 complex activities, “walk 2 miles or 3 kilometers” and “participate in recreation and

sports as you would like,” added in 1995^{24,25}. The 10 activities are scored without a calculator or computer, as a physical function (FN) score of 0–30, which may be recoded as 0–10 using a scoring template on Page 1. Three items (1k–1m) concerning sleep, anxiety, and depression have been found to be informative in patient care in the standard HAQ format²⁴, but are not scored formally.

The MDHAQ pain and global estimate VAS format is a 10-cm horizontal line format or 21 numbered circles. [At the time of these studies, the 10-cm horizontal line was used, but the 21-circle VAS is now used by each of the authors³⁶.] The RADAI self-report joint count^{30,34} is scored as 0–48; the raw 0–48 score may be recoded to 0–10 using a scoring template on Page 1. Boxes printed on the right side are included for the physician to record scores for pain, global estimate, and RADAI.

Page 2 of the MDHAQ (Figure 1B), the reverse side, includes a review of systems symptom checklist, scales for morning stiffness, change in status, exercise, fatigue VAS, recent medical history, and demographic data. All analyses in this report are derived from the scales on Page 1.

RAPID3 scores. RAPID3 scores are designed for usual clinical care, although they also may be useful for clinical research. The 3 Core Data Set measures on the MDHAQ, for function (FN), pain (PN), and patient global estimate (PTGL), are each scored 0–10 and recorded on the MDHAQ. The raw total score of 0–30 may be recoded to 0–10 using a scoring template at the bottom of Page 1. RAPID3 is mathematically identical to a patient activity score (PAS)²³. RAPID3 on the MDHAQ can be computed in about 10 seconds or less²⁶.

Other RAPID scores. Other RAPID scores that add further measures to RAPID3 were developed and analyzed to assess whether additional measures by a physician/assessor or patient might provide a substantially more informative index than RAPID3. Each index is labeled with a number after “RAPID” indicating the number of included measures, followed by abbreviations of these measures (Table 1).

RAPID4MDJC (Table 1) adds to RAPID3 a standard 28 swollen and tender joint count³⁷ performed by a physician/assessor, based on a rationale that this joint count is the most specific³⁸ and most highly valued³⁹ measure to assess patients with RA. To calculate RAPID4MDJC, the 28-joint count is scored 0–54 (0–28 tender joints; 0–26 swollen joints, not including the shoulder), recoded to a 0–10 scale using division by 5.4, then added to RAPID3 for a total of 0–40.

RAPID4PTJC (Table 1) adds to RAPID3 a RADAI self-report joint count³⁰, based on a rationale that a formal quantitative joint count is not performed at most RA patient visits¹¹, and a RADAI self-report joint count is correlated significantly with a physician/assessor joint count³⁰. As noted, the RADAI is scored 0–48, and recoded to a 0–10 scale using a scoring template on the MDHAQ. A raw RAPID4MDJC or RAPID4PTJC 0–40 scores may be divided by 4 to give an adjusted 0–10 score, using a template at the bottom of Page 1 (Figure 1)²⁶.

RAPID5 (Table 1) adds to RAPID4PTJC a physician/assessor global estimate, based on a rationale to include in an index the measure with the highest relative efficiency in most clinical trials^{15,40}, physician/assessor estimate of global status, as well as a joint count measure. The RAPID5 0–50 raw score may be divided by 5 to give an adjusted 0–10 score using a template at the bottom of Page 1 of the MDHAQ (Figure 1), computed in about 20 seconds²⁶.

Each of 2 to 5 measures included in a RAPID score is weighted equally on a 0–10 scale, in contrast to ACR improvement criteria²², DAS28³, and CDAI²⁸, in which joint-count data are weighted more heavily than other Core Set measures. Adjustment of all RAPID scores to 0–10 facilitates simple comparisons of all indices to one another and to DAS28 and CDAI. In usual care, RAPID3 may be scored 0–30, for further simplicity.

Other RAPID scores — such as RAPID2, which includes only a physician and a patient global estimate; RAPID4 versions that include RAPID3 plus a swollen joint count, tender joint count or physician global estimate; and RAPID5, with a physician joint count — were also computed. However, results were quite similar to RAPID3, RAPID4PTJC, RAPID4MDJC, and RAPID5, and are not presented in this report.

This questionnaire includes information not available from blood tests, X-rays, or any source other than you. Please try to answer each question, even if you do not think it is related to you at this time. Try to complete as much as you can yourself, but if you need help, please ask. There are no right or wrong answers. Please answer exactly as you think or feel. Thank you.

1. Please check (✓) the ONE best answer for your abilities at this time:

OVER THE LAST WEEK, were you able to:

Without ANY Difficulty	With SOME Difficulty	With MUCH Difficulty	UNABLE <u>To Do</u>

- a. Dress yourself, including tying shoelaces and doing buttons?
- b. Get in and out of bed?
- c. Lift a full cup or glass to your mouth?
- d. Walk outdoors on flat ground?
- e. Wash and dry your entire body?
- f. Bend down to pick up clothing from the floor?
- g. Turn regular faucets on and off?
- h. Get in and out of a car, bus, train, or airplane?
- i. Walk two miles or three kilometers, if you wish?
- j. Participate in recreational activities and sports as you would like, if you wish?
- k. Get a good night's sleep?
- l. Deal with feelings of anxiety or being nervous?
- m. Deal with feelings of depression or feeling blue?

0	1	2	3
0	1	2	3
0	1	2	3
0	1	2	3
0	1	2	3
0	1	2	3
0	1	2	3
0	1	2	3
0	1	2	3
0	1	2	3
0	1.1	2.2	3
0	1.1	2.2	3
0	1.1	2.2	3

1.04 FN (0-10)

1=0.3	16=5.3
2=0.7	17=5.7
3=1.0	18=6.0
4=1.3	19=6.3
5=1.7	20=6.7
6=2.0	21=7.0
7=2.3	22=7.3
8=2.7	23=7.7
9=3.0	24=8.0
10=3.3	25=8.3
11=3.7	26=8.7
12=4.0	27=9.0
13=4.3	28=9.3
14=4.7	29=9.7
15=5.0	30=10

2.PN (0-10)

4.PTGL (0-10)

RAPID 3 (0-30)

3.а-рПТJT(0-10)

1=0.2	25=5.2
2=0.4	26=6.4

4-0.8	28-5.8
5-1.0	29-6.0

7-1.5	31-6.4
8-1.7	32-6.7

10-2.1	34-7.1
11-2.3	35-7.3
12-2.4	36-7.4

14=2.9	38=7.9
15=3.1	39=8.1

17=3.5	41=8.5
18=3.8	42=8.8
19=4.1	43=9.1
20=4.4	44=9.4
21=4.7	45=9.7
22=5.0	46=10.0
23=5.3	47=10.3
24=5.6	48=10.6
25=5.9	49=10.9
26=6.2	50=11.2
27=6.5	51=11.5
28=6.8	52=11.8
29=7.1	53=12.1
30=7.4	54=12.4
31=7.7	55=12.7
32=8.0	56=13.0
33=8.3	57=13.3
34=8.6	58=13.6
35=8.9	59=13.9
36=9.2	60=14.2
37=9.5	61=14.5
38=9.8	62=14.8
39=10.1	63=15.1
40=10.4	64=15.4
41=10.7	65=15.7
42=11.0	66=16.0
43=11.3	67=16.3
44=11.6	68=16.6
45=11.9	69=16.9
46=12.2	70=17.2
47=12.5	71=17.5
48=12.8	72=17.8
49=13.1	73=18.1
50=13.4	74=18.4
51=13.7	75=18.7
52=14.0	76=19.0
53=14.3	77=19.3
54=14.6	78=19.6
55=14.9	79=19.9
56=15.2	80=20.2
57=15.5	81=20.5
58=15.8	82=20.8
59=16.1	83=21.1
60=16.4	84=21.4
61=16.7	85=21.7
62=17.0	86=22.0
63=17.3	87=22.3
64=17.6	88=22.6
65=17.9	89=22.9
66=18.2	90=23.2
67=18.5	91=23.5
68=18.8	92=23.8
69=19.1	93=24.1
70=19.4	94=24.4
71=19.7	95=24.7
72=20.0	96=25.0
73=20.3	97=25.3
74=20.6	98=25.6
75=20.9	99=25.9
76=21.2	100=26.2
77=21.5	
78=21.8	
79=22.1	
80=22.4	
81=22.7	
82=23.0	
83=23.3	
84=23.6	
85=23.9	
86=24.2	
87=24.5	
88=24.8	
89=25.1	
90=25.4	
91=25.7	
92=26.0	
93=26.3	
94=26.6	
95=26.9	
96=27.2	
97=27.5	
98=27.8	
99=28.1	
100=28.4	

21=4.4	45=9.4
22=4.6	46=9.6

24=5.0 48=10
RAPID 4 (0.4

[illegible]

MDGLOBAL(0-1)

100

RAPID 3 (03)

2. How much pain have you had because of your condition OVER THE PAST WEEK?

Please indicate below how severe your pain has been:

NO PAIN 0 0.5 1.0 1.5 2.0 2.5 3.0 3.5 4.0 4.5 5.0 5.5 6.0 6.5 7.0 7.5 8.0 8.5 9.0 9.5 10 PAIN AS BAD AS IT COULD BE

3. Please place a check (✓) in the appropriate spot to indicate the amount of pain you are having today in each of the joint areas listed below:

	None	Mild	Moderate	Severe		None	Mild	Moderate	Severe
<u>a.LEFT FINGERS</u>	☐ 0	☐ 1	☐ 2	☐ 3	<u>i.RIGHT FINGERS</u>	☐ 0	☐ 1	☐ 2	☐ 3
<u>b.LEFT WRIST</u>	☐ 0	☐ 1	☐ 2	☐ 3	<u>j.RIGHT WRIST</u>	☐ 0	☐ 1	☐ 2	☐ 3
<u>c.LEFT ELBOW</u>	☐ 0	☐ 1	☐ 2	☐ 3	<u>k.RIGHT ELBOW</u>	☐ 0	☐ 1	☐ 2	☐ 3
<u>d.LEFT SHOULDER</u>	☐ 0	☐ 1	☐ 2	☐ 3	<u>l.RIGHT SHOULDER</u>	☐ 0	☐ 1	☐ 2	☐ 3
<u>e.LEFT HIP</u>	☐ 0	☐ 1	☐ 2	☐ 3	<u>m.RIGHT HIP</u>	☐ 0	☐ 1	☐ 2	☐ 3
<u>f.LEFT KNEE</u>	☐ 0	☐ 1	☐ 2	☐ 3	<u>n.RIGHT KNEE</u>	☐ 0	☐ 1	☐ 2	☐ 3
<u>g.LEFT ANKLE</u>	☐ 0	☐ 1	☐ 2	☐ 3	<u>o.RIGHT ANKLE</u>	☐ 0	☐ 1	☐ 2	☐ 3
<u>h.LEFT TOES</u>	☐ 0	☐ 1	☐ 2	☐ 3	<u>p.RIGHT TOES</u>	☐ 0	☐ 1	☐ 2	☐ 3
<u>q.NECK</u>	☐ 0	☐ 1	☐ 2	☐ 3	<u>r.BACK</u>	☐ 0	☐ 1	☐ 2	☐ 3

4. Considering all the ways in which illness and health conditions may affect you at this time, please indicate below how you are doing:

VERY WELL 0 0.5 1.0 1.5 2.0 2.5 3.0 3.5 4.0 4.5 5.0 5.5 6.0 6.5 7.0 7.5 8.0 8.5 9.0 9.5 10 POORLY

Please turn to the other side

For Office Use Only: RAPID 3 RAPID 3 (0-10) RAPID 4 RAPID 4 (0-10)

NR: 1=0.3, 2=0.7, 3=1.0 LS: 4=1.3, 5=1.7, 6=2.0
MS: 7=2.3, 8=2.7, 9=3.0, 10=3.3, 11=3.7, 12=4.0
HS: 13=4.3, 14=4.7, 15=5.0, 16=5.3, 17=5.7, 18=6.0, 19=6.3, 20=6.7, 21=7.0, 22=7.3, 23=7.7, 24=8.0, 25=8.3, 26=8.7, 27=9.0, 28=9.3, 29=9.7, 30=10.0

NR: 1=0.3, 2=0.5, 3=0.8, 4=1.0 LS: 5=1.3, 6=1.5, 7=1.8, 8=2.0
MS: 9=2.3, 10=2.5, 11=2.8, 12=3.0, 13=3.3, 14=3.5, 15=3.8, 16=4.0
HS: 17=4.3, 18=4.5, 19=4.8, 20=5.0, 21=5.3, 22=5.5, 23=5.8, 24=6.0, 25=6.3, 26=6.5, 27=6.8, 28=7.0, 29=7.3, 30=7.5, 31=7.8, 32=8.0, 33=8.3, 34=8.5, 35=8.7, 36=9.0, 37=9.3, 38=9.5, 39=9.8, 40=10.0

NR: 1=0.2, 2=0.4, 3=0.6, 4=0.8, 5=1.0 LS: 6=1.2, 7=1.4, 8=1.6, 9=1.8, 10=2.0
MS: 11=2.2, 12=2.4, 13=2.6, 14=2.8, 15=3.0, 16=3.2, 17=3.4, 18=3.6, 19=3.8, 20=4.0
HS: 21=4.2, 22=4.4, 23=4.6, 24=4.8, 25=5.0, 26=5.2, 27=5.4, 28=5.6, 29=5.8, 30=6.0, 31=6.2, 32=6.4, 33=6.6, 34=6.8, 35=7.0, 36=7.2, 37=7.4, 38=7.6, 39=7.8, 40=8.0, 41=8.2, 42=8.4, 43=8.6, 44=8.8, 45=9.0, 46=9.2, 47=9.4, 48=9.6, 49=9.8, 50=10.0

RAPID 5 (0-10)

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Figure 1A. Multidimensional Health Assessment Questionnaire (MDHAQ), Version R783. The front page (A) includes 5 scales to assess physical function, psychological distress, pain, patient global estimate, and a self-report joint count on a RA Disease Activity Index (RADAI)^{30,34}. The 10 physical function activities (items 1a–1j) are each scored 0, 1, 2, or 3 (as with the HAQ), for a total of 0–30; the raw 0–30 score is recoded as 0–10 using a scoring template on the right side of the page. A brief psychological distress scale of 3 queries concerning sleep, anxiety, and depression (items 1k–1m) is given below the 10 activities; these queries may be informative to the rheumatologist in patient care, but are not scored formally. Scoring templates for pain, self-report joint count, and patient global estimate measures are also available on the right side of the page, and for RAPID indices at the bottom of the page. RAPID3 includes the 3 RA Core Data Set patient self-report measures: physical function, pain, and patient global estimate of status. RAPID4PTJC adds to RAPID3 a RADAI joint count, and RAPID5 adds to RAPID4PTJC a physician global estimate of patient status. Reprinted with permission: Health Report Services.

5. Please check (✓) if you have experienced any of the following over the last month:

☐ Fever	☐ Lump in your throat	☐ Paralysis of arms or legs
☐ Weight gain (>10 lbs)	☐ Cough	☐ Numbness or tingling of arms or legs
☐ Weight loss (<10 lbs)	☐ Shortness of breath	☐ Fainting spells
☐ Feeling sickly	☐ Wheezing	☐ Swelling of hands
☐ Headaches	☐ Pain in the chest	☐ Swelling of ankles
☐ Unusual fatigue	☐ Heart pounding (palpitations)	☐ Swelling in other joints
☐ Swollen glands	☐ Trouble swallowing	☐ Joint pain
☐ Loss of appetite	☐ Heartburn or stomach gas	☐ Back pain
☐ Skin rash or hives	☐ Stomach pain or cramps	☐ Neck pain
☐ Unusual bruising or bleeding	☐ Nausea	☐ Use of drugs not sold in stores
☐ Other skin problems	☐ Vomiting	☐ Smoking cigarettes
☐ Loss of hair	☐ Constipation	☐ More than 2 alcoholic drinks per day
☐ Dry eyes	☐ Diarrhea	☐ Depression - feeling blue
☐ Other eye problems	☐ Dark or bloody stools	☐ Anxiety - feeling nervous
☐ Problems with hearing	☐ Problems with urination	☐ Problems with thinking
☐ Ringing in the ears	☐ Gynecological (female) problems	☐ Problems with memory
☐ Stuffy nose	☐ Dizziness	☐ Problems with sleeping
☐ Sores in the mouth	☐ Losing your balance	☐ Sexual problems
☐ Dry mouth	☐ Muscle pain, aches, or cramps	☐ Burning in sex organs
☐ Problems with smell or taste	☐ Muscle weakness	☐ Problems with social activities

6. When you awakened in the morning OVER THE LAST WEEK, did you feel stiff? ☐ No ☐ Yes
If "No," please go to Item 7. If "Yes," please indicate the number of minutes _____, or hours _____ until you are as limber as you will be for the day.

7. How do you feel TODAY compared to ONE WEEK AGO? Please check (✓) only one.
Much Better ☐ (1), Better ☐ (2), the Same ☐ (3), Worse ☐ (4), Much Worse ☐ (5) than one week ago

8. How often do you exercise aerobically (sweating, increased heart rate, shortness of breath) for at least one-half hour (30 minutes)? Please check (✓) only one.
☐ 3 or more times a week (3) ☐ 1-2 times per month (1)
☐ 1-2 times per week (2) ☐ Do not exercise regularly (0) ☐ Cannot exercise due to disability/ handicap (9)

9. How much of a problem has UNUSUAL fatigue or tiredness been for you OVER THE PAST WEEK?
FATIGUE IS ☐ 0 ☐ 0.5 ☐ 1.0 ☐ 1.5 ☐ 2.0 ☐ 2.5 ☐ 3.0 ☐ 3.5 ☐ 4.0 ☐ 4.5 ☐ 5.0 ☐ 5.5 ☐ 6.0 ☐ 6.5 ☐ 7.0 ☐ 7.5 ☐ 8.0 ☐ 8.5 ☐ 9.0 ☐ 9.5 ☐ 10 FATIGUE IS A
NO PROBLEM MAJOR PROBLEM

10. Over the last 6 months have you had: [Please check (✓)]

☐ No ☐ Yes An operation	☐ No ☐ Yes Change(s) of arthritis drugs or other drugs
☐ No ☐ Yes Inpatient hospitalization	☐ No ☐ Yes Change(s) of address
☐ No ☐ Yes A new illness, accident or trauma	☐ No ☐ Yes Change(s) of marital status
☐ No ☐ Yes An important new symptom	☐ No ☐ Yes Change job or work duties, quit work, retired
☐ No ☐ Yes Side effect(s) of any drug	☐ No ☐ Yes Change of medical insurance, Medicare, etc.
☐ No ☐ Yes Smoke cigarettes regularly	☐ No ☐ Yes Change of primary care or other doctor

Please explain any "Yes" answer below, or indicate any other health matter that affects you:

SEX: ☐ Female, ☐ Male **ETHNIC GROUP:** ☐ Asian, ☐ Black, ☐ Hispanic, ☐ White, ☐ Other _____

Your Occupation _____ **Please circle the number of years of school you have completed:**

Work Status: ☐ Full-time, ☐ Part-time, ☐ Disabled 1 2 3 4 5 6 7 8 9 10

☐ Homemaker, ☐ Self-Employed, ☐ Retired, 11 12 13 14 15 16 17 18 19 20

☐ Seeking work, ☐ Other _____ **Please write your weight: _____ lbs. height: _____ inches**

Your Name _____ **Date of Birth** _____ **Today's Date** _____

Page 2 of 2 Thank you for completing this questionnaire to help keep track of your medical care. R783NP2

FOR OFFICE USE ONLY: I have reviewed [check(✓)]: _____ FN, _____ PN, _____ GL, _____ RADAI, _____ RAPID, _____ ROS, _____ HX

Name: _____ **Signature:** _____

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Figure 1B. The reverse side (B) includes a review of systems, morning stiffness, change in status, exercise activity, fatigue VAS, recent medical history, and demographic data. This information is not included in RAPID scores, but provides useful data for clinical care. Reprinted with permission: Health Report Services.

Table 1. Composition of Routine Assessment of Patient Index Data (RAPID) indices.

Measure	DAS28	CDAI	RAPID 3	RAPID 4PTJC	RAPID 4MDJC	RAPID 5
Number of measures included	4	4	3	4	4	5
Physical function			✓	✓	✓	✓
Pain			✓	✓	✓	✓
Patient global estimate	✓	✓	✓	✓	✓	✓
MD/Assessor global estimate		✓				✓
Tender joint count (MD)	✓	✓			✓	
Swollen joint count (MD)	✓	✓				
Patient joint count (RADAI)				✓		✓
ESR/CRP	✓					
Scale — raw score	0–10	0–76	0–30	0–40	0–40	0–50
Scale — adjusted score			0–10	0–10	0–10	0–10

MD: physician; RADAI: Rheumatoid Arthritis Disease Activity Index; ESR: erythrocyte sedimentation rate; CRP: C-reactive protein.

DAS28 and CDAI. DAS28³ includes 4 measures: 28 swollen joint count, 28 tender joint count, ESR, and patient global estimate, and is scored 0–10 using a DAS calculator (also available at the DAS website: <http://www.das-score.nl/www.das-score.nl/>). The CDAI^{12,28} includes 4 measures, 3 identical to DAS28, substituting a physician/assessor global estimate for ESR. The CDAI is scored as a simple 0–76 total: 0–28 for 2 joint counts, and 0–10 for 2 global estimates. Four DAS28 activity categories²⁷ are: > 5.1 = high, 3.21–5.1 = moderate, 2.61–3.2 = low, and ≤ 2.6 = remission. The corresponding CDAI categories²⁹ are > 22 = high, 10.1–22.0 = moderate, 2.9–10.0 = low, and ≤ 2.8 = remission. Proposed severity (rather than activity) categories for RAPID3 are: > 4 = high, 2.01–4 = moderate, 1.01–2 = low, and ≤ 1 = near-remission, on an adjusted 0–10 scale. On an unadjusted 0–30 scale, the severity categories are defined as > 12 = high, 6.01–12 = moderate, 3.01–6 = lower, and ≤ 3 = near-remission. These cutpoints were selected on the basis of clinical experience of the senior author over 20 years⁴¹ and analyses of adalimumab⁴² and abatacept²¹ clinical trial data.

Statistical analyses. The SPERA protocol was completed in 318 patients in the 3 clinical settings. One or more measures were missing in 33 patients (10.4%), primarily ESR and self-report of RADAI joint count. Analyses were conducted only in the 285 patients for whom complete data were available for all included measures and indices.

All data were entered into a Microsoft Access database, which had been developed for management of longitudinal studies³¹, and SPERA data in the cross-sectional, multinational QUEST-RA protocol³². The data were transferred to SAS® V9.2 (SAS, Cary, NC, USA) for statistical analyses. Demographic measures, clinical measures, RA Core Data Set measures, indices, and therapies in the 3 settings were compared using analysis of variance for continuous variables, and chi-square analysis for discontinuous variables. Spearman rank-nonparametric correlation coefficients were computed to compare individual Core Data Set measures, duration of disease, DAS28, CDAI, and various RAPID scores. Cross-tabulations were computed to compare 4 DAS28 and CDAI categories of high disease activity (DAS = 5.1–10; CDAI = 22–78), moderate activity (DAS = 3.2–5.1; CDAI = 10.1–22), low activity (DAS = 2.61–3.2; CDAI = 3.81–10), and remission (DAS = 0–2.6; CDAI = 0–28) to the 4 proposed RAPID categories of high severity (4.01–10), moderate severity (2.01–4.0), low severity (1.01–2.0), and near-remission (0–1.0). Statistical significance of the level of agreement of the different scales was evaluated using chi-square, kappa, and weighted kappa statistics.

RESULTS

Patients. The 285 patients included 88, 119, and 78 from 3 different clinical settings who had complete data available

(Table 2). The mean age was 57.4 years, 73% were female, 68.4% Caucasian, 18.6% African American, and 6.7% Hispanic. The mean duration of disease was 4.9, 8.2, and 14.0 years in practices established in 2001, 1987, and 1980, respectively. One practice included 50% African American patients and 23% Hispanic patients, who were a small minority in the other practices.

Among RA Core Data Set measures, mean patient questionnaire scores for function, pain, and global estimate did not differ significantly in the 3 practices, yielding mean RAPID3 scores of 2.9 (means of 2.7, 2.8, and 3.3 in the 3 practices). Overall mean swollen joint count was 3.7, tender joint count 3.5, physician global estimate 2.0, and ESR 23.4. Some differences were statistically significant, but clinically plausible, explained by lower swollen joint counts, lower physician global estimate, and higher ESR in one setting, and lower tender joint counts in another setting.

The mean DAS28 score of 3.4 differed statistically across the 3 settings, reflecting lower tender joint counts in one setting; no significant differences were seen between CDAI or RAPID scores. The overall mean DAS28 of 3.4, CDAI of 10.6, and RAPID3 of 2.9 are only slightly above the cutpoints for “low activity” or “low severity” in these 3 practices, in which an aggressive approach to control inflammation as completely as possible is pursued, although 17% of patients had high disease activity by DAS28 or CDAI (see below).

Overall, 61.8% of the patients were treated with prednisone, 72.3% with weekly low-dose methotrexate, and 28.4% with biological agents. In one practice, prednisone was taken by most patients, but with a mean dose of 4 mg per day, on a longterm basis⁴³. The data concerning these 285 patients appear to reflect a relatively typical population of patients with RA in the US, although variation was seen in many variables across the 3 settings, as seen in multicenter clinical trials.

Spearman correlations of DAS28, CDAI, and RAPID scores.

Table 2. Demographic, RA Core Data Set, indices, and medication data in 285 patients seen by 3 rheumatologists (MB, TP, YY) by setting. Values are mean (standard deviation) unless otherwise indicated.

	MB	TP	YY	All Patients	p
N	88	119	78	285	
Age, yrs	56.2 (15.6)	58.5 (14.5)	57.1 (13.5)	57.4 (14.6)	0.536
Duration, yrs	8.2 (6.8)	14.0 (10.1)	4.9 (6.0)	9.7 (9.0)	< 0.001
Education, yrs	13.6 (2.1)	13.8 (2.9)	13.4 (4.0)	13.6 (3.0)	0.604
Female*	62 (70.5)	82 (68.9)	64 (82.1)	208 (73.0)	0.076**
Race					< 0.001**
Caucasian*	68 (77.3)	113 (95.0)	14 (17.9)	195 (68.4)	
African American*	10 (11.4)	4 (3.4)	39 (50.0)	53 (18.6)	
Hispanic*	1 (1.1)	0	18 (23.1)	19 (6.7)	
RA Core Data Set measures					
Physician/assessor measures					
Swollen 28 joint count	4 (4.4)	4.3 (3.9)	2.4 (3.8)	3.7 (4.1)	0.004
Tender 28 joint count	1.5 (3.3)	4.4 (6)	4.3 (4.8)	3.5 (5.2)	< 0.001
Physician global estimate VAS	2.3 (1.8)	2.1 (1.1)	1.6 (1.4)	2.0 (1.4)	0.003
Laboratory measure					
Erythrocyte sedimentation rate	18.9 (22)	20.9 (18.5)	32.3 (33.2)	23.4 (24.9)	< 0.001
Patient measures					
Function	2.0 (1.8)	2.2 (1.9)	2.3 (2.0)	2.1 (1.9)	0.551
Pain VAS	3.2 (2.6)	3.3 (2.6)	4.0 (2.9)	3.5 (2.7)	0.104
Patient global estimate VAS	2.8 (2.4)	2.9 (2.4)	3.6 (2.7)	3.1 (2.5)	0.074
Clinical indices					
DAS28	2.7 (1.6)	3.6 (1.5)	3.7 (1.7)	3.4 (1.7)	< 0.001
CDAI	10.7 (9.6)	13.7 (10.8)	11.9 (11.1)	12.3 (10.6)	0.127
RAPID3	2.7 (2.1)	2.8 (2.1)	3.3 (2.4)	2.9 (2.2)	0.130
RAPID4PTJC	2.4 (1.9)	2.6 (2)	3.1 (2.2)	2.7 (2)	0.180
RAPID4MDJC	2.3 (1.7)	2.5 (1.8)	2.8 (2)	2.5 (1.8)	0.064
RAPID5	2.4 (1.8)	2.5 (1.7)	2.8 (2)	2.5 (1.8)	0.282
Treatment					
Prednisone*	39 (44.3)	108 (90.8)	29 (37.2)	176 (61.8)	< 0.001**
Methotrexate*	62 (70.5)	95 (79.8)	49 (62.8)	206 (72.3)	0.030**
Biologic agents*†	29 (33.0)	37 (31.1)	15 (19.2)	81 (28.4)	0.103**

* Values are reported as number of patients (percentage of total patients in column). Significance of differences between sites; ** p values for discontinuous variables were calculated by chi-square; all other p values (continuous variables) were calculated by analysis of variance (ANOVA). † Biologic agents include adalimumab, etanercept, and infliximab. VAS: visual analog scale.

Among the 285 patients with complete data, DAS28 was correlated significantly with RAPID3 ($\rho = 0.66$, $p < 0.001$; Table 3) and with other RAPID scores ($\rho = 0.65$ – 0.73 , $p < 0.001$; Table 3). CDAI was also correlated significantly with RAPID3, at somewhat higher levels than DAS28 ($\rho = 0.74$, $p < 0.001$; Table 3), as well as with all other RAPID scores ($\rho = 0.74$ – 0.83 , $p < 0.001$; Table 3). As expected,

the highest correlation was seen between DAS28 and CDAI ($\rho = 0.84$, weighted kappa = 0.60, $p < 0.001$), as 3 of the 4 measures in the DAS and CDAI (swollen joint count, tender joint count, patient global estimate) are identical. RAPID4MDJC, which includes 2 measures in common with DAS28 and CDAI (tender joint count, patient global estimate), was correlated at higher levels with DAS28 and CDAI than RAPID3, which includes only one measure found on the DAS28 and CDAI (patient global estimate). Nonetheless, RAPID3 was correlated significantly (at levels almost as high as RAPID scores with more common measures) with DAS28 and CDAI. The correlation of RAPID3 with DAS28 ($\rho = 0.66$) is greater than the correlation of CRP with ESR ($\rho = 0.51$) or of any Core Data Set measure with duration of disease (all $\rho < 0.16$).

Four categories of DAS28, CDAI, and RAPID3 scores. Among the 285 patients, 50 (17%) met DAS28 criteria for high activity (> 5.1), compared to 90 (32%) with moderate activity (3.3–5.1), 40 (14%) with low activity (2.7–3.2), and

Table 3. Spearman correlation coefficients for DAS28, CDAI, and all RAPID indices in 285 patients in 3 clinical settings.

	DAS28	CDAI	RAPID3	RAPID 4PTJC	RAPID 4MDJC
CDAI	0.844				
RAPID3	0.658	0.742			
RAPID4PTJC	0.654	0.748	0.989		
RAPID4MDJC	0.731	0.828	0.988		
RAPID5	0.692	0.805	0.981	0.991	0.985

$p < 0.001$ for all comparisons.

105 (37%) in remission (≤ 2.6) (Table 4A). The proportions of patients in 4 categories for RAPID3 were 31% with high severity (> 4.0), 25% with moderate severity (2.1–4.0), 18% with low severity (1.1–2.0), and 26% in near-remission (≤ 1.0) (Table 4A).

Among the 50 patients with high activity according to DAS28, 96% had high or moderate severity according to RAPID3 (Table 4A). Of the 105 patients in DAS28 remission, 73% had low severity or near-remission according to RAPID3 (Table 4A). The weighted kappa statistic for agreement of RAPID3 with DAS28 was 0.44 ($p < 0.001$).

CDAI criteria for high activity (> 22.0) were met by 17% of the patients, compared to 32% with moderate activity (10.1–22.0), 33% with low activity (2.9–10.0), and 18% with remission (≤ 2.8) (Table 4B). Among 50 patients with CDAI high activity, 96% had high or moderate RAPID3 severity (Table 4B). Among 52 patients in CDAI remission, 98% were in near-remission or low severity (Table 4B). The weighted kappa statistic for agreement of RAPID3 with CDAI was 0.51 ($p < 0.001$), somewhat higher than for DAS28.

Four categories of DAS28, CDAI, and other RAPID scores. Agreement of DAS28 (Table 5) and CDAI (Table 6) with RAPID4 and RAPID5 indices was quite similar to agreement with RAPID3. Of the 50 patients with DAS28 high activity, 94%–96% met high or moderate severity criteria for the other 3 RAPID scores (Table 5), similar to 96% for RAPID3 (Table 4A). Of the 105 patients in DAS28 remission, 76%–81% were in near-remission or low severity

according to the other RAPID scores (Table 5), similar to the 73% according to RAPID3 (Table 4A).

Of the 50 patients with CDAI high activity, 96%–98% met high or moderate severity criteria for the other RAPID scores (Table 6), similar to 96% for RAPID3 (Table 4B). Of the 52 patients in CDAI remission, 98%–100% were in RAPID near-remission or low severity categories (Table 6), similar to 98% for RAPID3 (Table 4B). Chi-square and weighted kappa statistics of 0.43–0.57 for comparisons of RAPID indices were all statistically significant ($p < 0.001$) and in the same range.

Two categories of DAS28, CDAI, and RAPID scores. Agreement between DAS, CDAI, and RAPID scores is summarized in Table 7 according to 2 categories: moderate and high activity/severity versus remission/near-remission and low activity/severity. The results suggest little incremental value to calculate indices that include additional measures beyond the 3 Core Data Set patient measures in RAPID3, particularly considering the time required.

DISCUSSION

Quantitative measures, ranging from blood pressure to serum glucose, have advanced clinical care in many diseases. Specific evidence of the value of quantitative data according to DAS28 has been documented in clinical trials of patients with RA^{4–9}. However, DAS28, or the simplified CDAI, requires a formal quantitative joint count, and is not available at most visits of patients with RA to a rheumatologist. A clinician can provide good patient care for RA

Table 4. RAPID3 scores compared to DAS28 and CDAI in 285 patients at 3 sites. All percentages are row percentages, except total in rightmost column (column percentages).

A. DAS28 vs RAPID3					
DAS28	RAPID3 Scores				Total
	4.1–10, High Severity	2.1–4.0, Moderate Severity	1.1–2.0, Low Severity	0–1.0, Near-remission	
> 5.1, high activity	37 (74%)	11 (22%)	1 (2%)	1 (2%)	50 (17%)
3.21–5.1, moderate activity	39 (43%)	27 (30%)	16 (18%)	8 (9%)	90 (32%)
2.61–3.2, low activity	4 (10%)	15 (38%)	10 (25%)	11 (27%)	40 (14%)
0–2.6, Remission	10 (10%)	18 (17%)	24 (23%)	53 (50%)	105 (37%)
Total	90 (31%)	71 (25%)	51 (18%)	73 (26%)	285
Kappa 0.26, weighted kappa 0.44.					
B. CDAI vs RAPID3					
CDAI	RAPID3 Scores				Total
	4.1–10, High Severity	2.1–4.0, Moderate Severity	1.1–2.0, Low Severity	0–1.0, Near-remission	
> 22, high activity	39 (78%)	9 (18%)	1 (2%)	1 (2%)	50 (17%)
10.1–22.0, moderate activity	36 (40%)	33 (36%)	15 (17%)	6 (7%)	90 (32%)
2.9–10, low activity	15 (16%)	28 (30%)	25 (27%)	25 (27%)	93 (33%)
0–2.8, remission	0 (0%)	1 (2%)	10 (19%)	41 (79%)	52 (18%)
Total	90 (31%)	71 (25%)	51 (18%)	73 (26%)	285
Kappa 0.32, weighted kappa 0.51.					

Table 5. DAS28 compared to other RAPID scores in 285 patients at 3 sites. All percentages are row percentages, except total in rightmost column (column percentages).

DAS28	RAPID4PTJC Scores				Total
	4.1–10, High Severity	2.1–4.0, Moderate Severity	1.1–2.0, Low Severity	0–1.0, Near-remission	
> 5.1 high activity	37 (74%)	10 (20%)	2 (4%)	1 (2%)	50 (17%)
3.21–5.1, moderate activity	30 (33%)	32 (36%)	20 (22%)	8 (9%)	90 (32%)
2.61–3.2, low activity	5 (12%)	14 (35%)	8 (20%)	13 (33%)	40 (14%)
0–2.6, remission	8 (7%)	18 (17%)	28 (27%)	51 (49%)	105 (37%)
Total	80 (28%)	74 (26%)	58 (20%)	73 (26%)	285

Kappa 0.26, weighted kappa 0.44.

DAS28	RAPID4MDJC Scores				Total
	4.1–10, High Severity	2.1–4.0, Moderate Severity	1.1–2.0, Low Severity	0–1.0, Near-remission	
> 5.1, high activity	37 (74%)	11 (22%)	2 (4%)	0 (0%)	50 (17%)
3.21–5.1, moderate activity	23 (25%)	41 (46%)	19 (21%)	7 (8%)	90 (32%)
2.61–3.2, low activity	1 (2%)	13 (33%)	15 (38%)	11 (27%)	40 (14%)
0–2.6, remission	3 (3%)	17 (16%)	29 (28%)	56 (53%)	105 (37%)
Total	64 (22%)	82 (29%)	65 (23%)	74 (26%)	285

Kappa 0.36, weighted kappa 0.53.

DAS28	RAPID5 Scores				Total
	4.1–10, High Severity	2.1–4.0, Moderate Severity	1.1–2.0, Low Severity	0–1.0, Near-remission	
> 5.1, high activity	37 (74%)	10 (20%)	3 (6%)	0 (0%)	50 (17%)
3.21–5.1, moderate activity	25 (28%)	38 (42%)	20 (22%)	7 (8%)	90 (32%)
2.61–3.2, low activity	4 (10%)	14 (35%)	12 (30%)	10 (25%)	40 (14%)
0–2.6, remission	5 (5%)	19 (18%)	27 (26%)	51 (51%)	105 (37%)
Total	71 (25%)	81 (28%)	62 (22%)	71 (25%)	285

Kappa 0.32, weighted kappa 0.48.

patients in most situations based on a history and qualitative physical examination, without quantitative data. However, availability of numerical data may enhance decisions, outcomes^{4–9}, and documentation of changes in patient status.

A careful history and physical examination, including a nonquantitative joint examination, form the foundation of any encounter of a physician and patient with RA. Nonetheless, RAPID3, which requires 10 seconds to calculate, can provide a valid index to supplement the findings with quantitative data in usual clinical care. Although correlations of RAPID with CDAI and DAS28 were highest for RAPID4MDJC, which includes a swollen and tender joint count by a physician/assessor, the incremental differences may not justify the 90 seconds required to perform a formal quantitative joint count as a routine practice.

It may appear inappropriate to suggest that a formal tender and swollen joint count performed by a physician/assessor is not required for an index to assess and monitor status of patients with RA in usual care. The joint examination

provides the primary information for diagnosis and monitoring of patients with RA and clearly reflects disease pathogenesis. A formal quantitative joint count is the most specific RA measure³⁹. However, several lines of evidence suggest that RAPID3, accompanied by a careful nonquantitative joint examination, but without a formal joint count, may have considerable value for usual RA care.

First, indices of only the 3 patient-reported Core Data Set measures are correlated with DAS28 in clinical trials of leflunomide^{18,19}, methotrexate^{18,19}, adalimumab²⁰, and abatacept²¹, and in clinical settings²³, and distinguish between active and control treatments in clinical trials as effectively as ACR and DAS criteria.

Second, the times required to score various RA measures include 90 seconds to perform a 28-joint count, 14.6 seconds to calculate DAS28 at the DAS website, 42 seconds to score a HAQ, 7.5 seconds to calculate 3 MDHAQ scores for physical function, pain and global status, 9.6 seconds to calculate RAPID3, and 20 seconds to calculate RAPID5²⁶. The

Table 6. CDAI compared to other RAPID scores in 285 patients at 3 sites. All percentages are row percentages, except total in rightmost column (column percentages).

CDAI	RAPID4PTJC Scores				Total
	4.1–10, High Severity	2.1–4.0, Moderate Severity	1.1–2.0, Low Severity	0–1.0, Near-remission	
> 22, high activity	38 (76%)	10 (20%)	1 (2%)	1 (2%)	50 (17%)
10.1–22.0, moderate activity	30 (33%)	35 (39%)	18 (20%)	7 (8%)	90 (32%)
2.9–10, low activity	12 (13%)	28 (30%)	30 (32%)	23 (25%)	93 (33%)
0–2.8, remission	0 (0%)	1 (2%)	9 (17%)	42 (81%)	52 (18%)
Total	80 (28%)	74 (26%)	58 (20%)	73 (26%)	285

Kappa 0.35, weighted kappa 0.52.

B. CDAI vs RAPID4MDJC

CDAI	RAPID4MDJC Scores				Total
	4.1–10, High Severity	2.1–4.0, Moderate Severity	1.1–2.0, Low Severity	0–1.0, Near-remission	
> 22, high activity	39 (78%)	10 (20%)	1 (2%)	0 (0%)	50 (17%)
10.1–22.0, moderate activity	22 (24%)	43 (48%)	21 (23%)	4 (5%)	90 (32%)
2.9–10, low activity	3 (3%)	29 (31%)	35 (38%)	26 (28%)	93 (33%)
0–2.8, remission	0 (0%)	0 (0%)	8 (15%)	44 (85%)	52 (18%)
Total	64 (22%)	82 (29%)	65 (23%)	74 (26%)	285

Kappa 0.42, weighted kappa 0.60.

C. CDAI vs RAPID5

CDAI	RAPID5 Scores				Total
	4.1–10, High Severity	2.1–4.0, Moderate Severity	1.1–2.0, Low Severity	0–1.0, Near-remission	
> 22, high activity	38 (76%)	10 (20%)	2 (4%)	0 (0%)	50 (17%)
10.1–22.0, moderate activity	25 (28%)	43 (48%)	18 (20%)	4 (4%)	90 (32%)
2.9–10, low activity	8 (9%)	28 (30%)	35 (38%)	22 (24%)	93 (33%)
0–2.8, remission	0 (0%)	0 (0%)	7 (13%)	45 (87%)	52 (18%)
Total	71 (25%)	81 (28%)	62 (22%)	71 (25%)	285

Kappa 0.42, weighted kappa 0.59.

Table 7. Agreement (percentage) between DAS, CDAI, and RAPID for moderate to high activity/severity versus remission/near-remission to low activity/severity.

Versus	DAS		CDAI	
	Moderate or High	Remission or Low	Moderate or High	Remission or Low
DAS	—	—	82%	83%
CDAI	82%	83%	—	—
RAPID3	81%	68%	84%	70%
RAPID4PTJC	78%	69%	81%	72%
RAPID4MDJC	80%	77%	81%	79%
RAPID5	79%	71%	83%	75%

time to score a CDAI may be similar to RAPID3. However, the 5–10 minutes to acquire the patient data for RAPID3 are those of a patient in the waiting room before seeing the physician, whereas the 90 seconds for a joint count to score a CDAI are taken from the interaction of physician and

patient during the limited time for this encounter. These differences may be important in efforts to incorporate quantitative measures into busy clinical settings.

Third, in part as a consequence of the time required, most rheumatologists do not perform a formal *quantitative* (non-quantitative) joint count in usual care, unless required for a clinical trial or medication¹¹, although most visits of patients with RA include a careful *qualitative* clinical examination. The RADAI self-report joint count is correlated with a tender joint count by a physician at levels of about $\rho = 0.5$ – 0.6 ³⁰, and $\rho = 0.53$ in the database in this study (data not shown). If an assessor is not available to perform a DAS28¹⁻³ or CDAI²⁸, a *quantitative* RAPID3 score and self-report joint count, along with a careful *qualitative* joint examination by a physician, may be sufficient for patient assessment and documentation in busy clinical settings.

Fourth, a formal quantitative joint count for swollen and tender joints performed by a physician/assessor has a num-

ber of limitations, which often are overlooked in the rheumatology literature¹⁰, including poor reliability⁴⁴⁻⁴⁸ (although reliability can be improved with training⁴⁶); lesser sensitivity to detect inflammation than magnetic resonance imaging (MRI) or ultrasound⁴⁹; greater improvement in patients who received placebo or control compared to active treatments than other Core Data Set measures⁴⁰; lesser prognostic value than physical function scores for important severe longterm outcomes such as work disability, costs, and mortality rates⁵⁰; and likelihood to be unchanged or improved over 5 years with traditional therapy while patients experienced progressive joint deformity and disability⁵¹. All reports emphasize that a joint count should be performed by the same observer at each visit. By contrast, it is possible to monitor a patient quantitatively using a patient questionnaire even if the usual rheumatologist is unavailable, if the patient sees another nonrheumatologist physician, or even at the patient's home or other settings.

We emphasize again that RAPID3 in no way is advocated to replace joint counts in clinical trials or a careful joint examination in clinical care. A formal quantitative joint count is appropriate for clinical trials and other clinical research, in which patients generally are described in groups, despite these limitations. Nonetheless, the limitations described detract from measurement accuracy in individual patients in busy clinical settings, in whom patient questionnaires provide greater reliability than joint counts⁵².

Although most patients who met criteria for moderate or high activity according to DAS28 or CDAI met criteria for moderate or high RAPID severity, a few patients had discrepant values. Perhaps these findings may be explained in part by sensitivity of patient questionnaires to longterm joint damage as well as inflammatory activity. Further analyses of discrepant scores are in progress (data not shown).

Several limitations are seen in this study. First, only 3 rheumatologists participated, and it would be desirable to extend these studies to a larger number of rheumatologists. Second, this was a cross-sectional study, and longitudinal data from clinical settings would appear desirable to study further the potential value of RAPID3 in helping to guide therapy. Rigorous longitudinal observations concerning RAPID3 are available from clinical trials¹⁸⁻²¹, and further longitudinal data from clinical care currently are being analyzed (data not shown).

The primary objective of this report is to document that categories of a RAPID3 score, an index that does not require a formal joint count, yield results generally quite similar to DAS28 and CDAI categories in usual clinical care, as also seen in clinical trials²¹. Addition of a physician/assessor joint count and/or physician global estimate resulted in somewhat higher correlations with DAS28 and CDAI, but added only marginally to classification of patients. RAPID3 is substantially more easily scored than DAS28 or CDAI, 10 seconds versus 90 seconds for a formal joint count.

Distribution of an MDHAQ to each patient at each visit in the infrastructure of usual clinical care has been the practice in the 3 clinical settings of the authors, for 25, 7, and 3 years, with completion of the questionnaire by > 99% of patients⁵³. This practice causes no disruption of patient flow, saves time for the rheumatologist, and provides far superior documentation of patient status than is available in usual rheumatology care. Further use of RAPID3 to assess, monitor, and document patient status quantitatively in busy clinical settings could improve care, enhance documentation, and lead to better outcomes for patients with rheumatic diseases and for the field of rheumatology.

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